FULL STACK DEVELOPMENT

PROJECT DOCUMENTATION

Streamlining IT Procurement Through Standard Laptop Order Automation

Team ID: SWTID-2026-7609

Team Member	Role
Bhavani V	Team Leader
Danushtiya V	Team Member
Dhanusha S	Team Member
Janani K	Team Member
Kavi Priya K	Team Member

Project Domain: IT Procurement / Workflow Automation

Technology Context: ServiceNow (Workflow Studio, Service Catalog, Approvals)

1. Introduction

Project Title: Streamlining IT Procurement Through Standard Laptop Order Automation

Team ID: SWTID-2026-7609

Team Members: Dharshinyanabhan KP (Team Leader) and Catherine Sweety R (Team Member).

This project provides a structured digital workflow for requesting and processing standard laptop orders. Employees order an approved laptop from a Service Catalog, the order is checked against a price-based approval rule, and, once approved, a fulfilment task is created automatically and assigned to the hardware team. The solution replaces scattered emails and manual records with a single, traceable request, approval and task trail.

2. Project Overview

Purpose: The purpose of the project is to standardize laptop procurement and make ordering faster, more transparent and easier to track. Users submit a structured request from the catalog, and a defined approval and fulfilment workflow handles the rest without manual hand-offs.

Key Features:

- Standard laptop catalog item (Standard Laptop, Lenovo Carbon X1, listed at $1,100) in the Hardware category of the Service Catalog.
- Single request form: the requester selects the item and submits it, which generates a Request (REQ) and a Requested Item (RITM).
- Approval step: high-value requests (total price above $1,000) are routed to an approver, who can approve or reject.
- Automation flow "Standard laptop tasks" built in Workflow Studio, triggered by the Service Catalog.
- Automatic creation of a Catalog Task with a short description, description and the Hardware assignment group.
- End-to-end traceability through linked records: REQ, RITM, approval record and SCTASK.
- Clear status values on every record (Pending Approval, Requested, Approved, Open) for tracking.

3. Architecture

The solution is built on the ServiceNow platform. The three layers requested by the template map to platform components as follows.

Frontend: The requester uses the ServiceNow Service Catalog (Hardware category) to browse and order the laptop, and then views the resulting Request, Requested Item, Approval and Catalog Task forms. The flow designer uses Workflow Studio to build and activate the automation.

Backend: The business logic lives in a Flow built in Workflow Studio named "Standard laptop tasks". Its trigger is *Service Catalog* and its action is *Create Catalog Task*, which receives the Requested Item record through a data pill. The approval rule and record handling are performed by the platform's server-side engine.

Database: Data is stored in standard ServiceNow tables and linked through reference fields.

Entity (Table)	Important Fields
Catalog Item (sc_cat_item)	Name, Category (Hardware), Catalogs, Price, Active, Process Engine (Flow)
Request (sc_request)	Number (REQ), Requested for, Opened by, Price, Approval, Request state
Requested Item (sc_req_item)	Number (RITM), Item, Quantity, Price, Approval
Approval (sysapproval_approver)	Approver, State, Approval reason, Comments, Created
Catalog Task (sc_task)	Number (SCTASK), Short description, Description, Assignment group, Request item, Approval, State

High-Level Flow: Employee → Service Catalog → Standard Laptop item → Request (REQ) → Approval → Flow "Standard laptop tasks" → Catalog Task (SCTASK) for the Hardware group → Fulfilment.

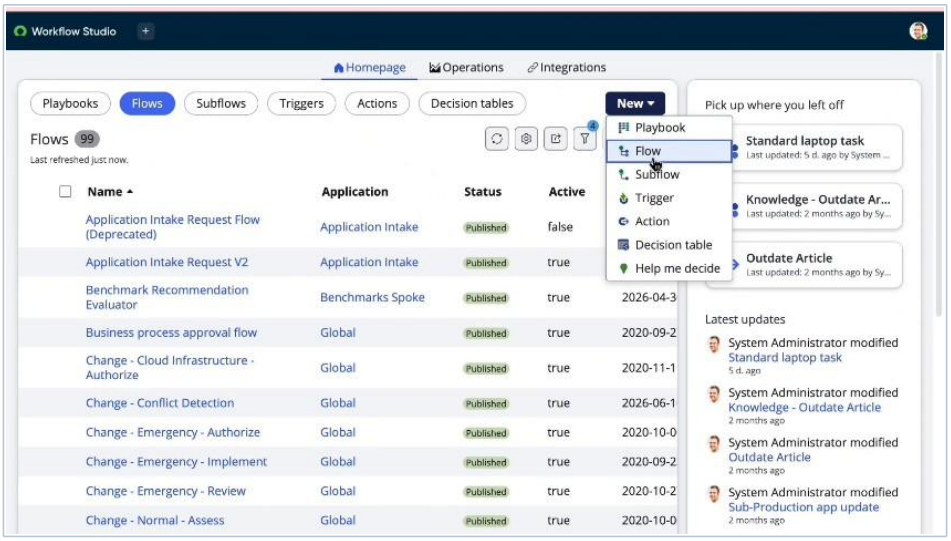


Figure 1: Workflow Studio home page; a new Flow is created from the New menu.

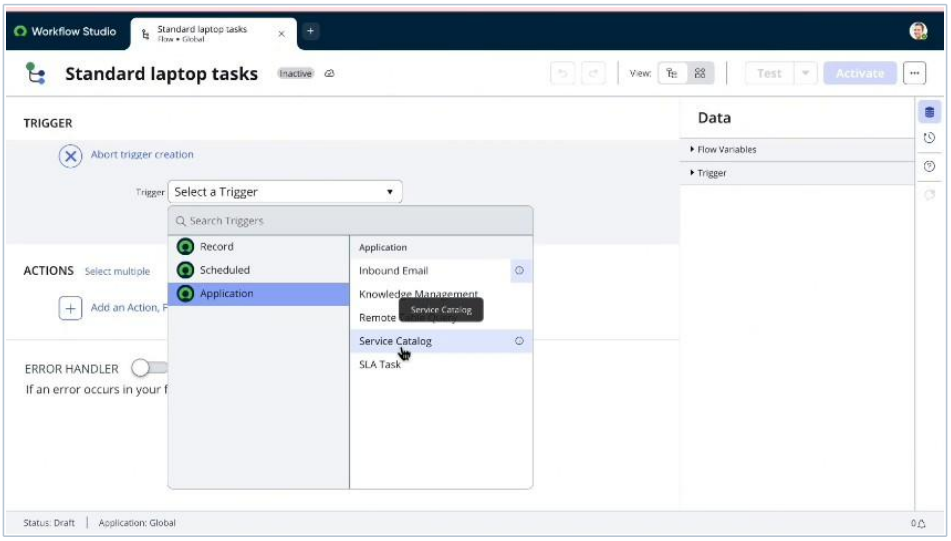


Figure 2: Trigger configuration: the Service Catalog trigger is selected under the Application category.

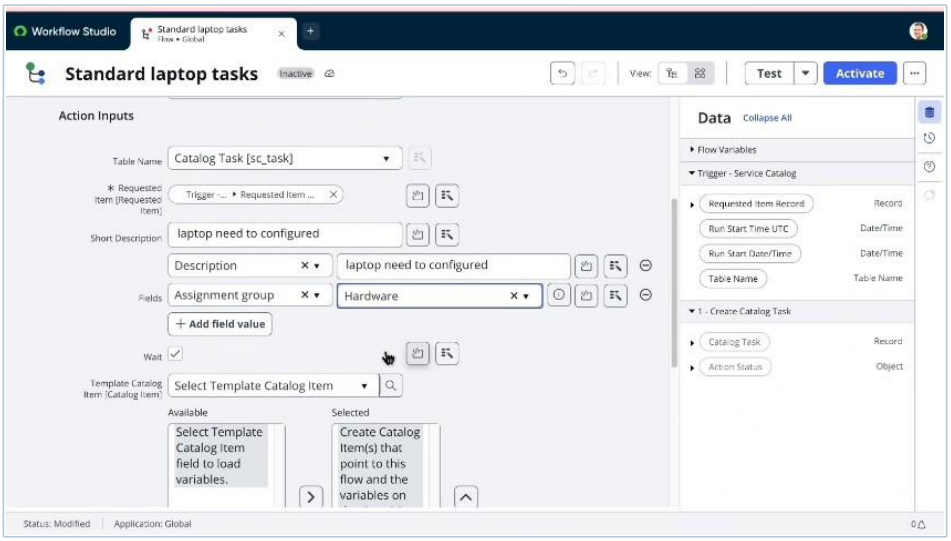


Figure 3: Create Catalog Task action: Description and the Hardware assignment group are set as field values.

4. Setup Instructions

Prerequisites: A modern web browser; a ServiceNow developer account with a Personal Developer Instance (PDI); and an account on the instance with the administrator role. No local installation, database or build tools are required because the platform is hosted.

Installation:

- Sign in to the ServiceNow Developer site and request a Personal Developer Instance.
- Open the instance (https://<instance-name>.service-now.com) and log in as the administrator.
- Confirm that the Service Catalog contains the Hardware category and the Standard Laptop catalog item (Catalog Items list).
- Open Workflow Studio and create the flow described in Section 3 (trigger: Service Catalog; action: Create Catalog Task).
- Set the action inputs: Requested Item (trigger data pill), Short Description, Description and Assignment group (Hardware).
- Link the flow to the catalog item from the Process Engine tab of the Standard Laptop item, then activate the flow (Section 6).

Instance credentials must be kept private and must not be shared in the project repository or report.

5. Folder Structure

ServiceNow is a hosted platform, so there is no client/server source folder. The equivalent organization of the solution is shown below.

Client (user-facing layer): Service Catalog > Hardware > Standard Laptop, and the Request, Requested Item, Approval and Catalog Task forms.

Server (logic and data layer): The Workflow Studio flow, the approval engine and the platform tables.

```
ServiceNow instance (Application scope: Global)
|-- Service Catalog
|    +-- Hardware
|         +-- Standard Laptop (sc_cat_item)  -> Process Engine: Flow
|-- Workflow Studio
|    +-- Flows
|         +-- Standard laptop tasks
|              |-- Trigger : Service Catalog
|              +-- Action 1: Create Catalog Task (sc_task)
+-- Records generated at run time
     |-- Request           REQ0010001    (sc_request)
     |-- Requested Item    RITM0010001   (sc_req_item)
     |-- Approval                        (sysapproval_approver)
     +-- Catalog Task      SCTASK0010001 (sc_task)
```

6. Running the Application

There are no npm commands for this solution. The workflow is started by activating the flow and then placing an order from the catalog.

- **Activate:** Open the flow in Workflow Studio and click Activate; confirm the prompt.
- **Order:** Go to Service Catalog > Hardware and order the Standard Laptop item.
- **Approve:** Sign in as the approver, open the approval record and click Approve.
- **Verify:** Open the Request and the generated Catalog Task to confirm the status.

Typical sequence: Create flow → Activate flow → Order laptop → Approval → Catalog Task created → Fulfilment.

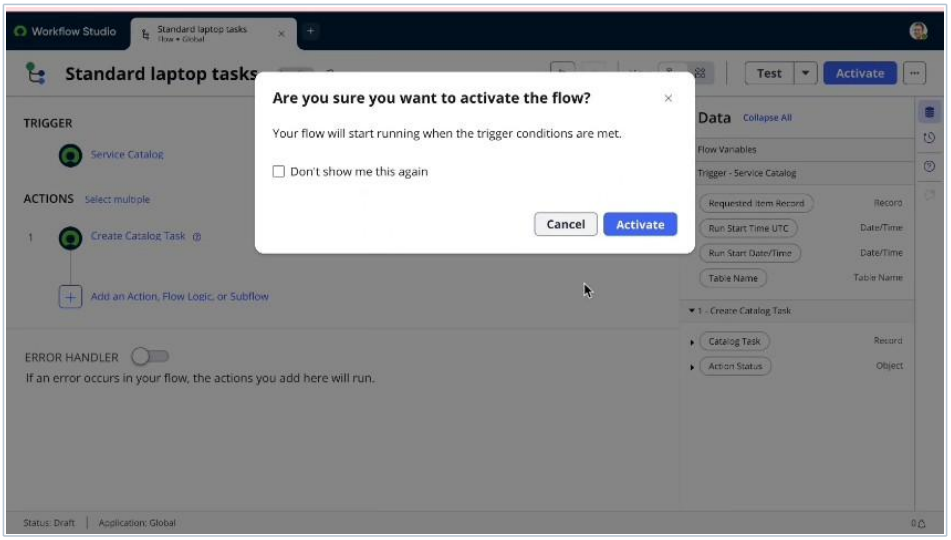


Figure 4: Activation prompt: the flow starts running when the trigger conditions are met.

7. API Documentation

The project is built with platform features and does not add custom endpoints. The ServiceNow REST APIs listed below are available on the instance for the same records and can be used for integrations. Requests need an authenticated user with suitable roles.

Method	Endpoint	Purpose
GET	/api/now/table/sc_cat_item	List catalog items such as Standard Laptop.
POST	/api/sn_sc/servicecatalog/items/{sys_id}/order_now	Order a catalog item (creates a request).
GET	/api/now/table/sc_request	Retrieve requests such as REQ0010001.
GET	/api/now/table/sc_req_item	Retrieve requested items (RITM).
GET	/api/now/table/sysapproval_approver	Retrieve approval records and their state.
PATCH	/api/now/table/sysapproval_approver/{sys_id}	Approve or reject by updating the state.
GET	/api/now/table/sc_task	Retrieve catalog tasks such as SCTASK0010001.

Example order request body: { "sysparm_quantity": "1" }. **Example response (illustrative):** { "result": { "number": "REQ0010001", "table": "sc_request" } }.

8. Authentication

Access is controlled by ServiceNow's built-in user authentication and role-based access control. Users sign in to the instance and the platform maintains the session; REST calls use authenticated credentials. What a user can do depends on the roles assigned:

- **System Administrator:** builds the flow, manages the catalog item and raises the request in the demonstration.
- **Approver (Eric Schroeder in the demonstration):** receives the approval record and approves or rejects.
- **Hardware group:** works the Catalog Task created by the flow.

Approval records move through the states Not Yet Requested, Requested, Approved and Rejected, as available in the flow's Approval field.

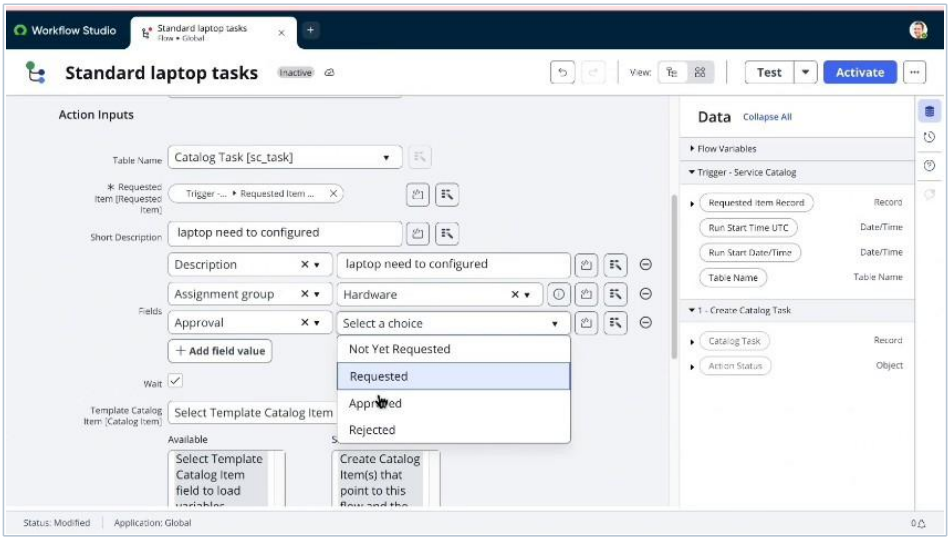


Figure 5: Approval field values available on the Create Catalog Task action.

9. User Interface

The interface follows a simple procurement journey, from the catalog to the approval and the resulting task.

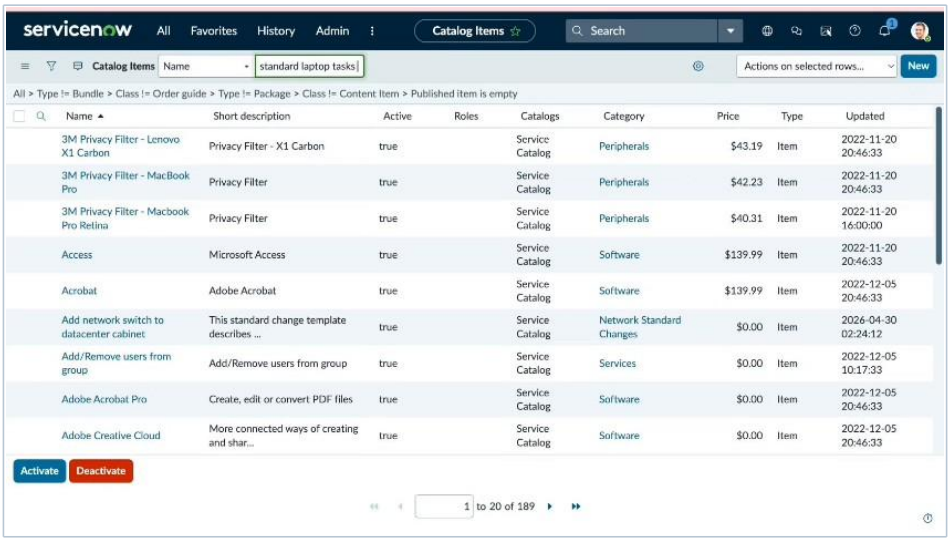


Figure 6: Catalog Items list (administrator view), used to locate the Standard Laptop item.

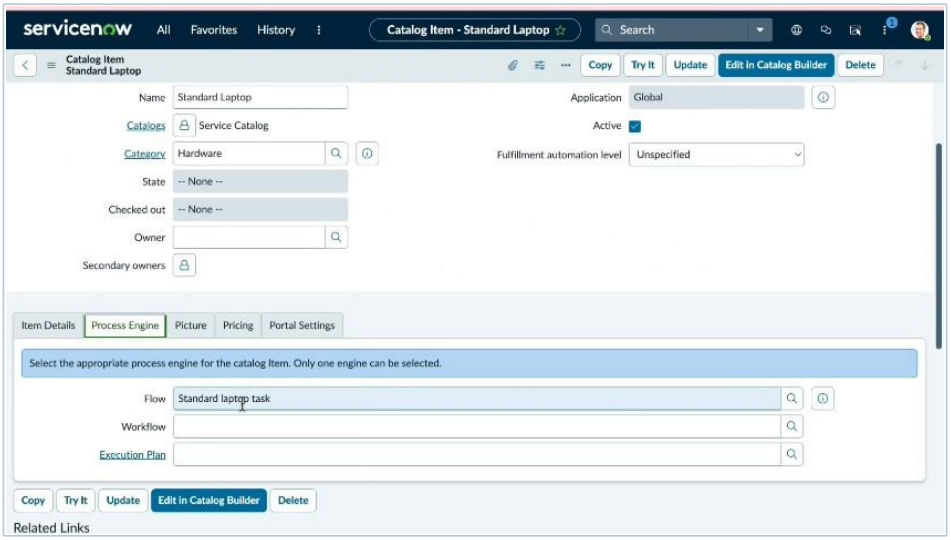


Figure 7: Standard Laptop catalog item (Hardware); the Process Engine tab links the item to the flow.

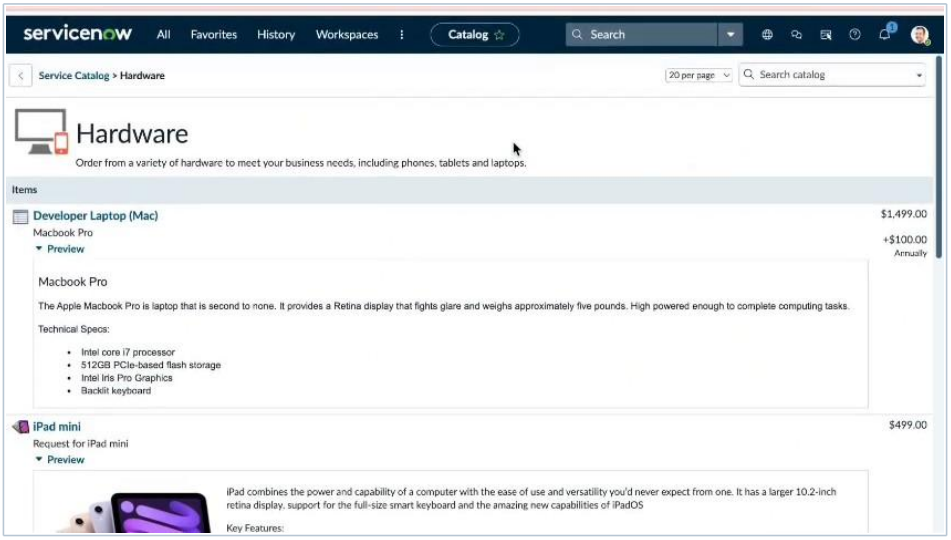


Figure 8: Service Catalog > Hardware page listing the orderable hardware items.

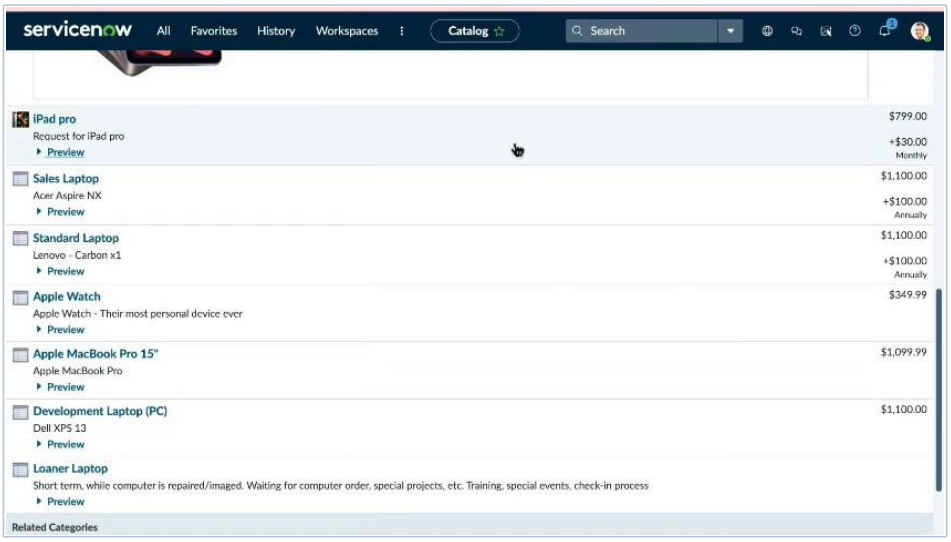


Figure 9: Standard Laptop (Lenovo Carbon X1) listed at $1,100 in the Hardware catalog.

10. Testing

Testing was carried out as manual, end-to-end functional testing on the developer instance, following one complete order through every stage of the workflow.

#	Test case	Expected result	Result
1	Create and activate the flow	Flow is saved and activation is confirmed	Passed
2	Open the Hardware catalog	Standard Laptop is listed at $1,100	Passed
3	Order the Standard Laptop	Request REQ0010001 created, state Pending Approval	Passed
4	Check approval routing	Approval record for the approver in state Requested	Passed
5	Approve the request	Approval state changes to Approved	Passed
6	Check task creation	Catalog Task SCTASK0010001 is created and open	Passed

Further tests planned: rejection path, orders at or below the approval threshold, multiple quantities, and access checks for users without the approver role.

11. Screenshots or Demo

The demonstration follows one complete standard laptop order, as listed below. A screen recording of the run is available from the team.

- Build and activate the flow (Figures 1 to 4).
- Open the Hardware catalog and select Standard Laptop (Figures 8 and 9).
- Place the order and show the generated request number and its Pending Approval state (Figure 10).
- Show the approval record for the approver (Figure 11).
- Approve the request (Figure 12).
- Show the final approval status and the generated Catalog Task (Figures 13 and 14).

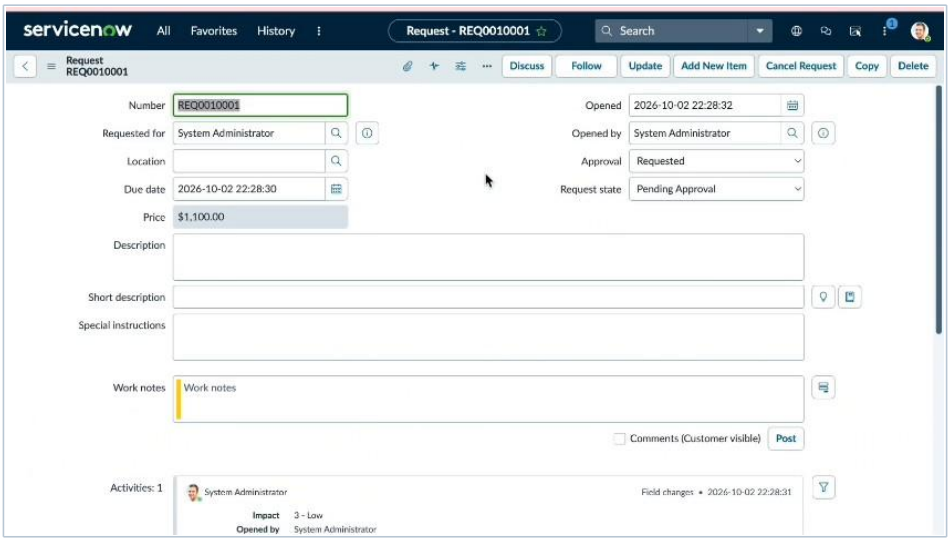


Figure 10: Request REQ0010001 for $1,100 with Approval = Requested and Request state = Pending Approval.

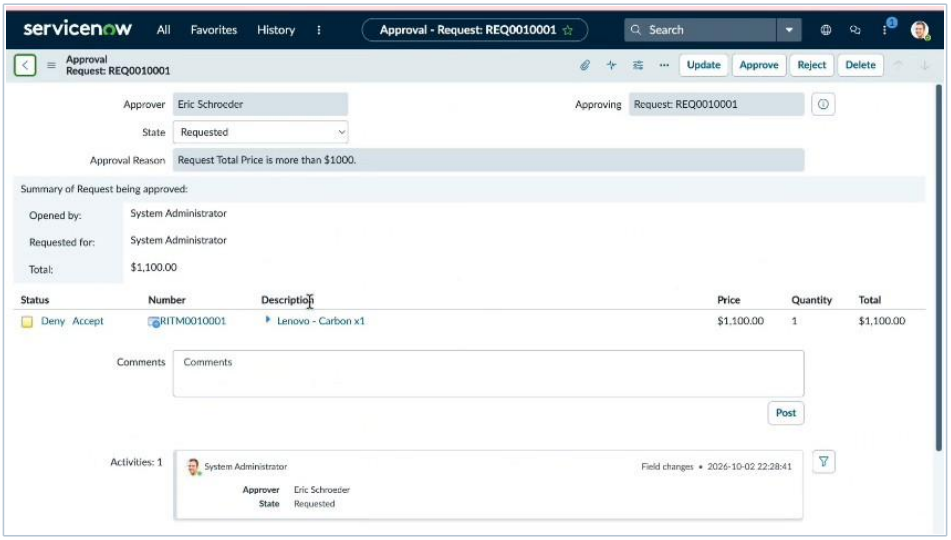


Figure 11: Approval record: the approver sees the reason (request total price is more than $1,000) and the item.

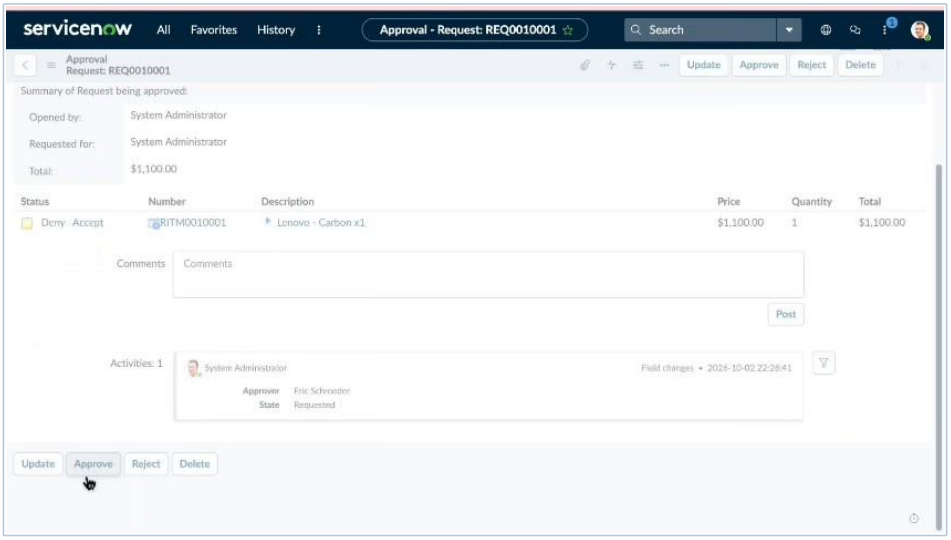


Figure 12: The approver approves the request using the Approve button.

12. Known Issues

- The solution depends on an active ServiceNow instance; a Personal Developer Instance can hibernate after inactivity and needs to be woken up.
- The flow currently creates one generic Catalog Task; catalog variables and template catalog items are not mapped to the task.
- The task text in the demonstration ("Laptop need to Configured") needs wording clean-up.
- Notifications (email or chat) were not set up or demonstrated in this version.
- The rejection path and orders below the approval threshold have not yet been demonstrated.

13. Future Enhancements

- Add inventory checks against the IT asset database before approving an order.
- Create purchase orders and integrate approved vendors.
- Add budget-limit validation and quotation comparison.
- Send email or Teams notifications on submission, approval and task assignment.
- Add dashboards for order volume, cost, processing time and fulfilment performance.
- Support multiple laptop models and role-based approval levels by department.
- Generate asset tags automatically when a laptop is assigned to an employee.

14. Results

The demonstration confirmed that the automated workflow runs end to end for a standard laptop order.

Item	Outcome
Catalog item	Standard Laptop (Lenovo Carbon X1), $1,100, quantity 1
Request	REQ0010001 created; initial state Pending Approval
Requested item	RITM0010001
Approval	Raised for the approver (reason: total price above $1,000); moved from Requested to Approved
Catalog Task	SCTASK0010001 created with Hardware assignment; Approval = Approved, State = Open

- The order, approval and task are linked, so the full history of one request can be traced from a single record.
- The approval rule prevented the task from being treated as approved until the approver acted.
- Fulfilment work is now routed automatically to the hardware team instead of being handled by email.

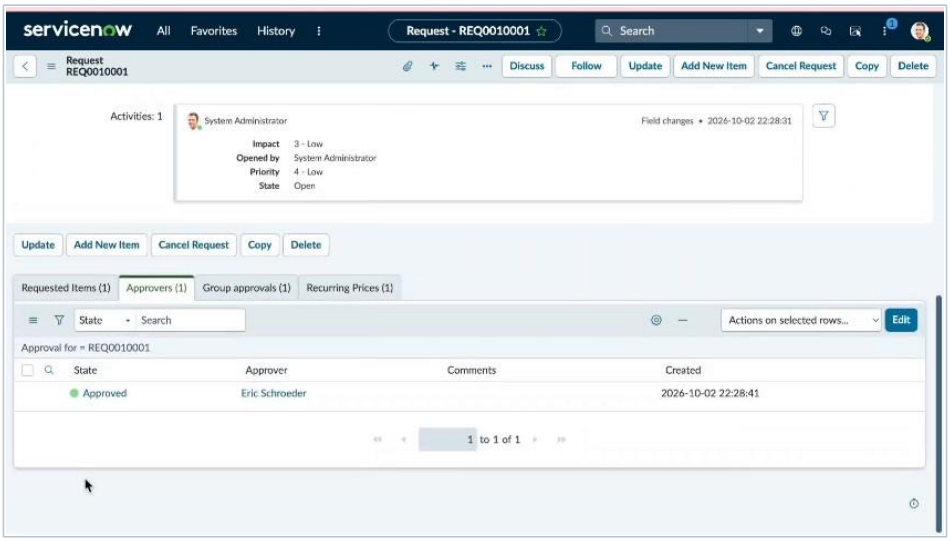


Figure 13: Approvers tab on the request: the approval by Eric Schroeder shows as Approved.

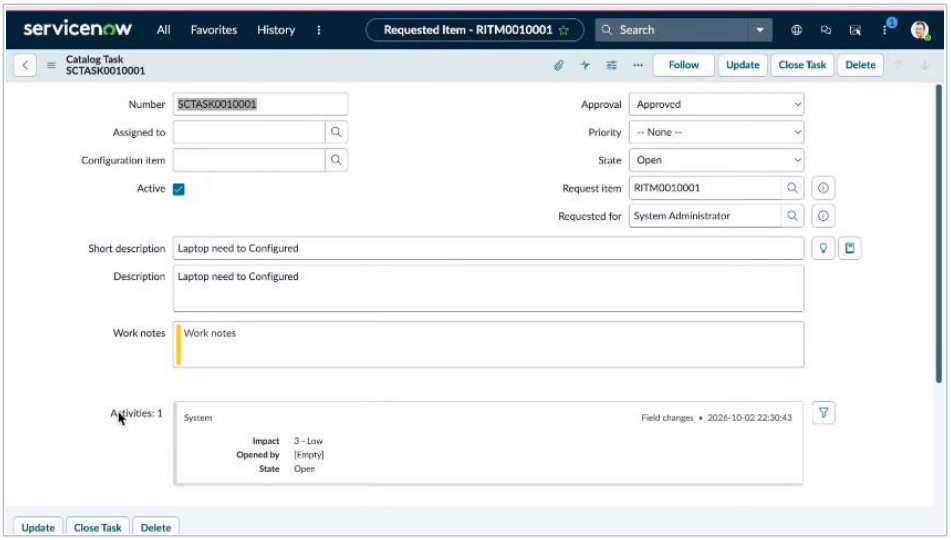
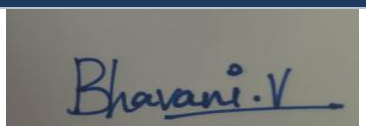
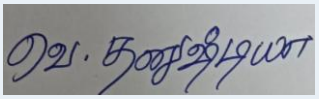
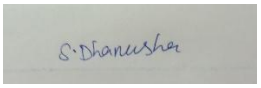
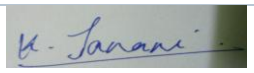
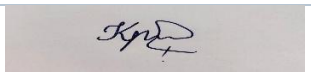


Figure 14: Catalog Task SCTASK0010001 generated with Approval = Approved and State = Open.

Sign-off

Team ID: **SWTID-2026-7609**. By signing below, the team confirms that this document accurately describes the project.

Name	Role	Signature	Date
Bhavani V	Team Leader		03 Oct 2026
Danushtiya V	Team Member		03 Oct 2026
Dhanusha S	Team Member		03 Oct 2026
Janani K	Team Member		03 Oct 2026
Kavi Priya K	Team Member		03 Oct 2026

End of Project Documentation